

Julien HORGUEDEBAT

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Presentation

Student in the Master program in Advanced Mathematics at the Université Marie et Louis Pasteur in Besançon (France)

Education

Université Marie et Louis Pasteur Sept 2023 – present
Master degree in Advanced Mathematics

- **Coursework:** Numerical Analysis, Partial and Ordinary Differential Equations, Student Seminar, Fields Theory, Algebra, Complex Analysis, Fourier Analysis, Topology, Probabilities

Université Marie et Louis Pasteur Sept 2022 – June 2023
Bachelor degree in Fundamental Mathematics

- Result: 15.415/20
- **Coursework:** Algebra, Metric spaces, Integration Theory, Functional Analysis, Probabilities, Affine Structure, Numerical Analysis, Differential calculus

Université Marie et Louis Pasteur Sept 2022 – present
Magistère in Mathematics

- **Coursework:** Epistemology, Statistics, Functional Analysis, Introduction to C, Colloquium

Experience

Research Internship Besançon, FR
Laboratoire de Mathématiques de Besançon - UMR 6623 May 2024 – Jul 2024

- Implementation of numerical methods to solve integral equations using the Nyström method.

Handler Besançon, FR
Grand Besançon Métropole June 2023

Seminars Attended

Laboratoire de Mathématiques de Besançon - UMR 6623 Sept 2025 – present
Numerical Analysis and Scientific Computing

- Weekly attending in the workgroup and seminar of the Numerical Analysis and Scientific Computing team of the laboratory.

Research Initiation School of the LmB Jan 2023
Winter School of the Maths Master of LmB Jan 2025

Laboratoire de Mathématiques de Besançon - UMR 6623 Dec 2025

- Conferences by members of the laboratory of mathematics

Lectures Sophie Kowalevski 4-6 June 2024
Centre Henri Lebesgue - Angers

- Short-courses in algebra and analysis/probabilities.

Academic Projects

Moment Problems Oct 2025 – present
Université Marie et Louis Pasteur; directed by Thierry DAUDE and Bastien POLIZZI

- Methods of resolution for moment problems and numerical implementation
- Tools: Julia, $\text{\LaTeX}$

Master 1 thesis: Fourier Transform on a Finite Commutative Group Jan 2024 – May 2024
Université Marie et Louis Pasteur; directed by Cécile ARMANA; group of 2

- Project about group theory, Fourier transform on finite commutative groups, Fast Fourier Transform and

applications to polynomial multiplication.

- Tools: Python, $\text{\LaTeX}$

Bachelor Project: On the existence of invariant measures under the action of a group and the Banach-Tarski paradox

Jan 2023 – May 2023

Université Marie et Louis Pasteur; directed by Marsault CHABAT; group of 3

- Project about group theory, the Banach-Tarski paradox and the existence of non-measurable sets in $\mathbb{R}$.
- Tools: $\text{\LaTeX}$

Skills

Programming languages: Python (Advanced), Julia (Advanced), $\text{\LaTeX}$ (Advanced), C (basic)

Languages: French (Mother Tongue), English (Technical), Spanish (Basic)

First aid certification (PSC1)